CHIMERA PLATFORM

Fractional Services Unit

Email Data Collection

chimera-fsu-email-ingest | v1.0

Project Strategy & Technical Specification

Prepared for	Mark Insley / Cape Berkshire Ltd
Prepared by	Charles — Lead Developer
Date	March 2026
Status	Active Development — Phase 1
FSU	chimera-fsu-email-ingest
Gmail	chimera.data.in@gmail.com

1. Executive Summary

The Email Data Collection FSU (Fractional Services Unit) is the first purpose-built intelligence ingestion module in the Chimera platform rebuild. It establishes the foundational data pipeline that will feed the platform's AI analytics layer — transforming unstructured email communications into structured, indexed, queryable intelligence.

> *An FSU is a standalone, independently deployable microservice with a single, well-defined domain function. Each FSU exposes a standard Chimera API, stores its data in the central Chimera registry, and is designed to plug into any Chimera agent or orchestration layer without bespoke integration work.*

The dedicated Gmail address chimera.data.in@gmail.com serves as the centralised intake point for all email-borne intelligence entering the Chimera system. Emails forwarded from any number of source accounts are automatically captured, parsed by content type, analysed by an AI model (Claude by Anthropic), enriched with structured metadata tags, and stored in Google Cloud Storage — with a queryable index maintained in Firestore.

The result is a searchable, agent-accessible knowledge base that grows automatically with every inbound email, requiring zero manual data entry.

2. Business Context & Strategic Rationale

2.1 The Problem

The Chimera platform processes a continuous stream of intelligence from multiple sources: racing data, market signals, strategy updates, and operational communications. Much of this intelligence arrives via email — but email is inherently unstructured, unsearchable, and inaccessible to automated systems without significant manual processing overhead.

Currently, valuable data points buried in emails require manual extraction, cross-referencing, and entry into the platform. This is slow, error-prone, and does not scale.

2.2 The Solution

The Email Data Collection FSU eliminates this bottleneck entirely. By treating the Gmail inbox as an automated data pipeline rather than a manual communications tool, the FSU transforms every inbound email into a structured intelligence record — automatically, in real time, with zero human intervention.

- Racing data, tips, and form guides forwarded from data providers are automatically parsed and indexed

- Strategy communications from Mark are captured, summarised, and tagged with action items
- Betfair reports and market data arrive as email — the FSU extracts and structures this data
- Alert emails from monitoring systems are classified by urgency and routed appropriately
- Any medium — text, PDF, image, audio — is processed and made searchable

2.3 Strategic Value

This FSU is the entry point for a broader Chimera intelligence architecture. Once email data is structured and indexed, it becomes immediately available to:

- The Chimera AI Analytics Agent — for pattern recognition and signal extraction
- The Reporting FSU — for inclusion in automated daily and weekly reports to Mark
- The Signal Intelligence Pipeline — as a data source alongside market feeds
- Future FSUs — via the standard Chimera inter-FSU API protocol

3. System Overview

3.1 How It Works — Plain English

The system operates as follows:

1. An email arrives in chimera.data.in@gmail.com — either sent directly or forwarded from another account.
2. Google Cloud Pub/Sub instantly notifies the Chimera FSU service (running on Google Cloud Run) that a new email has arrived. No polling — event-driven, real time.
3. The FSU fetches the full email via the Gmail API: headers, body text, HTML, and all attachments.
4. Attachments are routed by type: PDFs and Word documents have text extracted; images are passed through OCR; audio files are transcribed to text.
5. All raw content is stored immediately in Google Cloud Storage — nothing is lost.
6. Claude (Anthropic AI) analyses the email and produces structured intelligence metadata: a descriptive title, summary, topic tags, entities (people, organisations, race venues, horse names), intent classification, urgency rating, relevancy score, and action items.
7. The complete intelligence record is saved to the Chimera Data Registry (Firestore) — a structured, searchable index accessible to any Chimera agent or service.

3.2 Data Flow Diagram

EMAIL ARRIVES → Pub/Sub Notification → Gmail API Fetch → [Parse Body + Attachments] → [OCR / STT / Text Extract] → [Store Raw to GCS] → [AI Tagging via Claude] → [Store Processed Record] → [Write to Firestore Registry] → AVAILABLE TO AGENTS

3.3 What Gets Stored

Every email produces two storage artefacts:

- Raw Store (GCS): The original email content — body text, HTML, all attachments in original format, extracted texts, and transcripts. This is the permanent, immutable record.
- Intelligence Record (Firestore): The AI-generated metadata — title, summary, topics, entities, tags, intent, urgency, relevancy score. This is the searchable, queryable index.

4. Intelligence Metadata Schema

Every email processed by the FSU produces an intelligence record with the following structure. This is the data model that agents, dashboards, and reports consume.

4.1 Source Metadata

message_id	Unique Gmail message identifier — used for idempotency
thread_id	Gmail thread identifier — groups related emails
from_address	Sender email address
from_name	Sender display name
subject	Original email subject line
received_at	Timestamp of receipt (ISO 8601)
forwarded_from	Original account if email was forwarded
gmail_labels	Gmail labels applied at time of receipt

4.2 AI-Generated Intelligence

title	Descriptive title generated by Claude (max 80 chars)
summary	2-3 sentence plain-English summary of the email
topics	Array of topic tags (e.g. 'lay betting', 'form guide', 'stake management')

entities.people	Named individuals mentioned in the email
entities.organisations	Companies, bodies, and organisations mentioned
entities.race_venues	Horse racing venues identified
entities.horse_names	Horse names identified in the content
entities.monetary_values	Financial figures extracted (e.g. stake amounts, winnings)
intent	Email classification: informational | action_required | data_payload | alert | report
urgency	Urgency level: low | medium | high | critical
relevancy_score	0.0 to 1.0 score of relevance to Chimera operations
relevancy_reasoning	AI explanation of the relevancy score
chimera_domain_tags	Domain-specific Chimera tags (e.g. 'signal-intelligence', 'spread-control')
action_items	Extracted action items requiring follow-up
contains_pii	Boolean flag — whether email contains personal identifiable information
sentiment	Overall sentiment: neutral | positive | negative

5. Technical Architecture

5.1 Technology Stack

Backend Service	FastAPI (Python) — Cloud Run
Email Trigger	Gmail API + Google Cloud Pub/Sub (event-driven, not polling)
AI Intelligence Layer	Claude claude-sonnet-4-20250514 (Anthropic API)
Image OCR	Google Cloud Vision API
Audio Transcription	Google Cloud Speech-to-Text
PDF/Document Extract	PyMuPDF + python-docx
Data Registry	Google Cloud Firestore
Blob Storage	Google Cloud Storage
Secret Management	Google Secret Manager
CI/CD	Cloud Build → Cloud Run (auto-deploy on GitHub commit)
Monitoring	Google Cloud Logging + Cloud Monitoring
API Auth	X-Chimera-API-Key header (per-FSU key)

5.2 GCS Bucket Structure

chimera-email-data/ raw/{year}/{month}/{day}/{message_id}/ email_metadata.json ← full source metadata body.txt ← clean body text body.html ← original HTML body attachments/ ← original attachment files processed/{record_id}/ record.json ← complete intelligence record extracted_texts/ ← text extracted from attachments transcripts/ ← audio transcriptions index/ daily_manifest_{date}.json

5.3 Supported Media Types

Plain Text / HTML	Parsed directly — body text extracted and cleaned
PDF Documents	Text extracted via PyMuPDF — full content indexed
Word Documents (.docx)	Text extracted via python-docx — full content indexed
Images (JPEG, PNG, GIF)	OCR via Google Vision API — text and descriptions extracted
Audio (MP3, WAV, FLAC)	Transcribed via Google Speech-to-Text — transcript indexed
Excel / CSV	Stored raw — structured data extraction in roadmap
ZIP Archives	Stored raw — contents extraction in roadmap
Other	Stored raw with MIME type flagged for review

6. FSU API Reference

The FSU exposes a RESTful API following the Chimera API Convention. All protected endpoints require the X-Chimera-API-Key header.

6.1 System Endpoints

Method	Endpoint	Auth	Description
GET	`/health`	None	Cloud Run liveness probe
GET	`/status`	None	Operational status snapshot
GET	`/version`	None	FSU and API version info

6.2 Configuration Endpoints

Method	Endpoint	Auth	Description
GET	/v1/config	Required	Get current processing configuration
PUT	/v1/config	Required	Update processing parameters
GET	/v1/config/schema	Required	JSON schema for valid configuration

6.3 Ingest Endpoints

Method	Endpoint	Auth	Description
POST	/v1/ingest/pubsub-push	None	Cloud Pub/Sub push receiver — called by Google on new email
POST	/v1/ingest/manual	Required	Manually trigger processing for a specific Gmail message ID
POST	/v1/ingest/reprocess/{id}	Required	Re-run AI tagging on an existing record (e.g. after model upgrade)
GET	/v1/ingest/queue	Required	View records currently in pending status

6.4 Registry Endpoints

Method	Endpoint	Auth	Description
GET	/v1/registry	Required	Query registry — filter by topic, intent, urgency, domain tag, sender, relevancy, status
GET	/v1/registry/{record_id}	Required	Fetch complete intelligence record by ID
GET	/v1/registry/metrics	Required	Processing stats: totals, by status, by intent
POST	/v1/registry/agent/query	Required	AI agent integration — structured query for agent consumption

6.5 Sources Endpoints

Method	Endpoint	Auth	Description
GET	/v1/sources	Required	List all registered forwarding sources
POST	/v1/sources	Required	Register a new forwarding source
DELETE	/v1/sources/{source_id}	Required	Remove a forwarding source

7. Development Phases

The FSU is built incrementally across four phases. Each phase delivers a working, deployable unit — no phase leaves the system in a broken state.

Phase	Name	Deliverable	Dependency	Duration
Phase 1	Reliable Capture	Gmail watch + Pub/Sub trigger + raw GCS storage. Every email captured without loss.	GCP setup complete, Gmail API enabled	1-2 weeks
Phase 2	AI Intelligence	Email parser + Claude AI tagging + Firestore registry. Full intelligence records produced.	Phase 1 complete + Anthropic API key	1-2 weeks
Phase 3	FSU API Layer	Complete REST API: config management, registry query, agent endpoint, sources management.	Phase 2 complete	1 week
Phase 4	Attachment Pipelines	PDF/DOCX text extraction, image OCR, audio transcription. All media types fully processed.	Phase 3 complete + Vision/STT APIs enabled	1-2 weeks

7.1 Phase 1 — Reliable Capture

Goal: Get every email into GCS without loss. No AI yet. Prove the pipeline is reliable before adding intelligence.

- Create chimera.data.in@gmail.com and configure Gmail API access
- Set up Pub/Sub topic: chimera-email-ingest-trigger
- Deploy Cloud Run service with /v1/ingest/pubsub-push endpoint
- Implement Gmail watch() subscription on service startup
- Parse email headers and body, store raw artefacts to GCS
- Implement idempotency check to prevent duplicate processing
- Set up Cloud Build pipeline for auto-deploy on GitHub commit

7.2 Phase 2 — AI Intelligence

Goal: Every captured email gets a complete Claude-generated intelligence record in Firestore.

- Integrate Anthropic Claude API for AI tagging
- Implement EmailIntelligence schema with all metadata fields
- Configure Chimera domain hints for relevancy scoring
- Store complete intelligence records in Firestore registry
- Implement fallback tagging for AI failure scenarios
- Validate tagging quality against sample emails

7.3 Phase 3 — FSU API Layer

Goal: Full API surface live. Mark or any agent can query the registry, update config, and inspect the system.

- Implement all registry query endpoints with filter support
- Build config management endpoints with JSON schema validation
- Create sources management endpoints
- Implement agent/query endpoint for AI agent integration
- Add metrics endpoint
- Secure all endpoints with X-Chimera-API-Key authentication
- Write API documentation

7.4 Phase 4 — Attachment Pipelines

Goal: No email content left unprocessed. PDFs, images, audio — all indexed.

- PDF text extraction via PyMuPDF
- Word document text extraction via python-docx
- Image OCR via Google Cloud Vision API
- Audio transcription via Google Cloud Speech-to-Text
- Store all extracted texts and transcripts to GCS
- Link extracted content back to parent intelligence record

8. Gmail Account Setup Guide

8.1 Configuring Forwarding Sources

Set up forwarding from each source email account:

Gmail accounts	Settings → Forwarding → Add forwarding address → chimera.data.in@gmail.com → confirm verification
Outlook / Hotmail	Settings → Mail → Forwarding → enter chimera.data.in@gmail.com
Google Workspace	Admin panel mail routing or individual account forwarding rules
Custom domain email	Domain mail admin forwarding rule → chimera.data.in@gmail.com
Any SMTP client	Create a mail rule: forward [criteria] to chimera.data.in@gmail.com

Recommendation: Apply filters on source accounts before forwarding. Forward only emails matching relevant senders or subjects. Forwarding everything will create noise and inflate processing costs.

9. Chimera API Convention

The Chimera API Convention is the standard contract that every FSU must implement. It ensures that any agent, dashboard, or orchestrator can interact with any FSU without bespoke integration — today's email FSU, tomorrow's racing data FSU, the FSU after that.

9.1 FSU Naming

chimera-fsu-{domain}-{function}Examples: chimera-fsu-email-ingest | chimera-fsu-racing-data-feed | chimera-fsu-betfair-signal | chimera-fsu-reporting-engine

9.2 Standard Request Envelope

```
{  "chimera_version": "1.0",  "request_id": "uuid-v4",  "issued_at": "2026-03-04T10:00:00Z",
"issued_by": "agent-id | user-id | system",  "payload": {}}
```

9.3 Standard Response Envelope

{ "chimera_version": "1.0", "request_id": "echo-of-inbound-id", "fsu": "chimera-fsu-email-ingest", "status": "success | error | partial", "timestamp": "2026-03-04T10:00:00Z", "data": {}, "errors": [], "meta": { "processing_time_ms": 342, "version": "1.0.0" }}

9.4 Mandatory Endpoints

Every FSU, regardless of domain function, must implement these endpoints:

Method	Endpoint	Auth	Description
GET	/health	None	Cloud Run liveness probe
GET	/status	None	Operational status
GET	/version	None	FSU and API version
GET	/v1/config	Required	Current processing parameters
PUT	/v1/config	Required	Update processing parameters
GET	/v1/registry	Required	Query data register (paginated, filterable)
GET	/v1/registry/{id}	Required	Fetch single record
POST	/v1/registry/agent/query	Required	AI agent integration endpoint
GET	/v1/metrics	Required	Processing stats
POST	/v1/trigger	Required	Manual trigger / reprocess

10. Risks & Considerations

Gmail Watch Expiry	Gmail API watch() subscriptions expire every 7 days. The service must renew automatically — handled on startup and via a scheduled Cloud Scheduler job.
Pub/Sub Message Volume	If the inbox receives high volume, Cloud Run will scale instances. Min-instances: 1 ensures no cold start on the push endpoint.
AI Tagging Cost	Each email processed incurs an Anthropic API call. Implement the ignore_senders config to filter out known noise sources (newsletters, automated notifications).
PII in Emails	Emails may contain personal data. The contains_pii flag is set by Claude but is advisory. Review data retention policies if storing emails containing third-party PII.
Attachment Size	Large attachments (multi-MB PDFs, audio files) will increase processing time. Cloud Run timeout is set to 300 seconds — monitor for timeout events.

Gmail API Quotas	Gmail API has a daily quota. At normal operational email volume this is not a concern, but worth monitoring if the inbox receives bulk forwarding.
Secrets Management	API keys and service account credentials must be stored in Google Secret Manager only — never in environment variables or committed to GitHub.

11. Deployment

11.1 Ongoing Deployment

All commits to the main branch on GitHub automatically trigger Cloud Build, which builds the Docker container, pushes to Google Container Registry, and deploys the new version to Cloud Run with zero downtime. No manual deployment steps required after initial setup.

12. Future Roadmap

The Email Data Collection FSU is the first node in a growing Chimera FSU ecosystem. Planned future units that will consume and extend the email intelligence layer:

- chimera-fsu-racing-data-feed — Structured racing data ingestion from data providers
- chimera-fsu-betfair-signal — Real-time Betfair market signal processing
- chimera-fsu-reporting-engine — Automated daily/weekly reporting to Mark, drawing on email registry
- chimera-fsu-alert-router — Urgency-based alert routing and notification management
- chimera-fsu-analytics-agent — Cross-FSU AI analytics and pattern recognition

Planned enhancements to this FSU:

- Natural language query endpoint — ask the registry questions in plain English
- Excel and CSV structured data extraction
- ZIP archive unpacking and recursive content processing
- Email thread awareness — group related emails by thread with cumulative intelligence
- Duplicate content detection — identify when the same data arrives via multiple routes
- Scheduled daily manifest generation — summary of intelligence captured each day

13. Appendix — Glossary

FSU	Fractional Services Unit — a single-domain, independently deployable Chimera microservice
chimera.data.in	The dedicated Gmail address that serves as the centralised email intake point
Pub/Sub	Google Cloud Pub/Sub — the event messaging service that notifies the FSU of new emails
Cloud Run	Google's serverless container platform — hosts the FSU backend service
Firestore	Google's NoSQL document database — the Chimera Data Registry index
GCS	Google Cloud Storage — stores raw and processed email artefacts
Gmail watch()	Gmail API method that subscribes an inbox to Pub/Sub notifications
Intelligence record	The AI-generated metadata produced for each email — stored in Firestore
Domain hints	Keywords provided to Claude to calibrate relevancy scoring for Chimera's domain
Idempotency	The guarantee that processing the same email twice produces the same result without duplication
Chimera API Convention	The standard API contract that all FSUs must implement for consistent integration

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